

28

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$$q\Delta V = \text{gain in KE}$$

$$q\Delta V = p^2/2m - 0$$

$$\Rightarrow p = \sqrt{2mq\Delta V}$$

Using de Broglie's equation,

$$\lambda = h/p = h/\sqrt{2qm\Delta V}$$

Since m and q are the same for both cases, $\lambda \propto 1/\sqrt{V}$.

$$\lambda_0/\lambda_1 = \sqrt{100/1} = 10$$

29

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